



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO NAGUL ENERGY SUPPLYIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

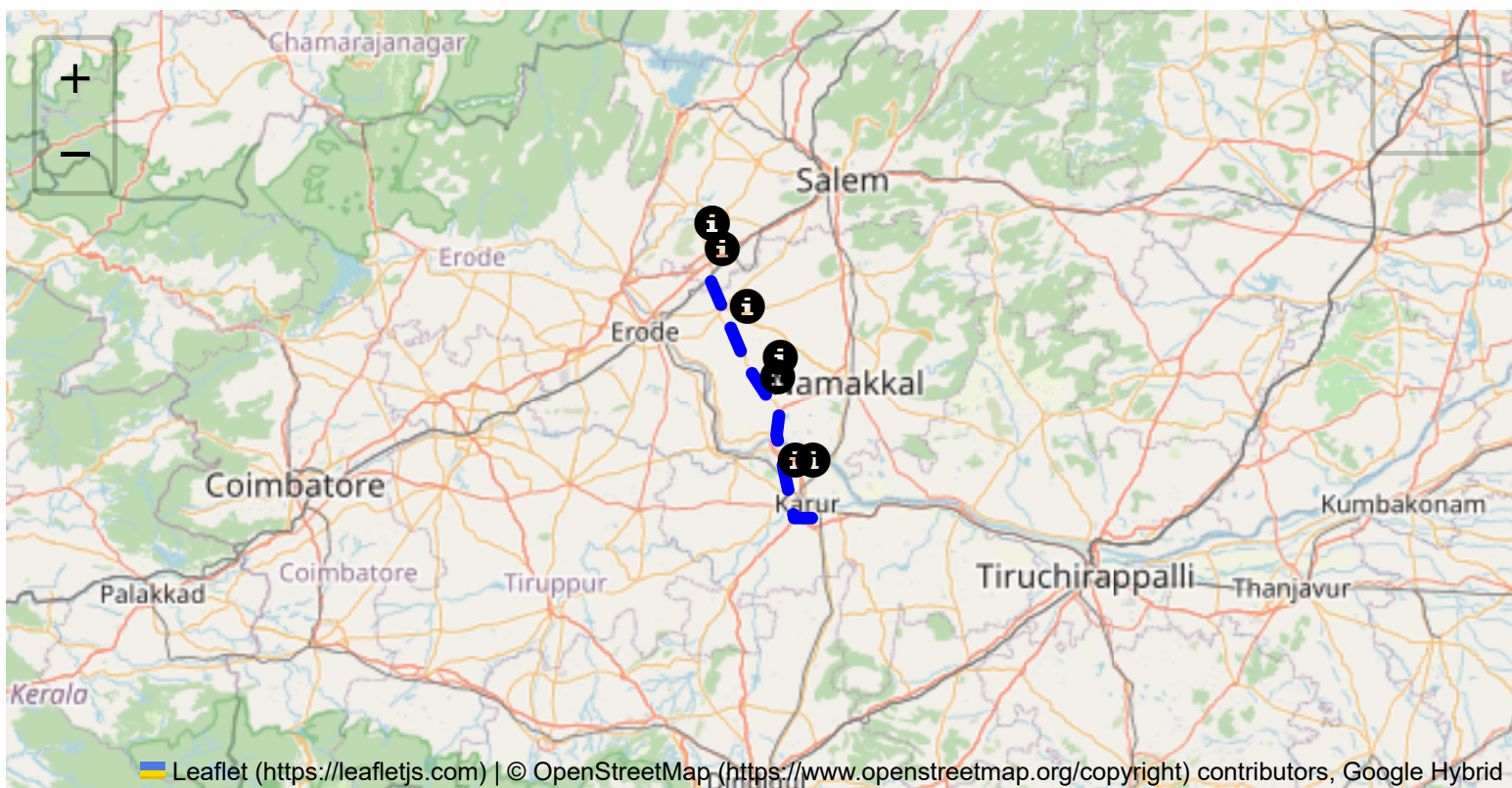
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

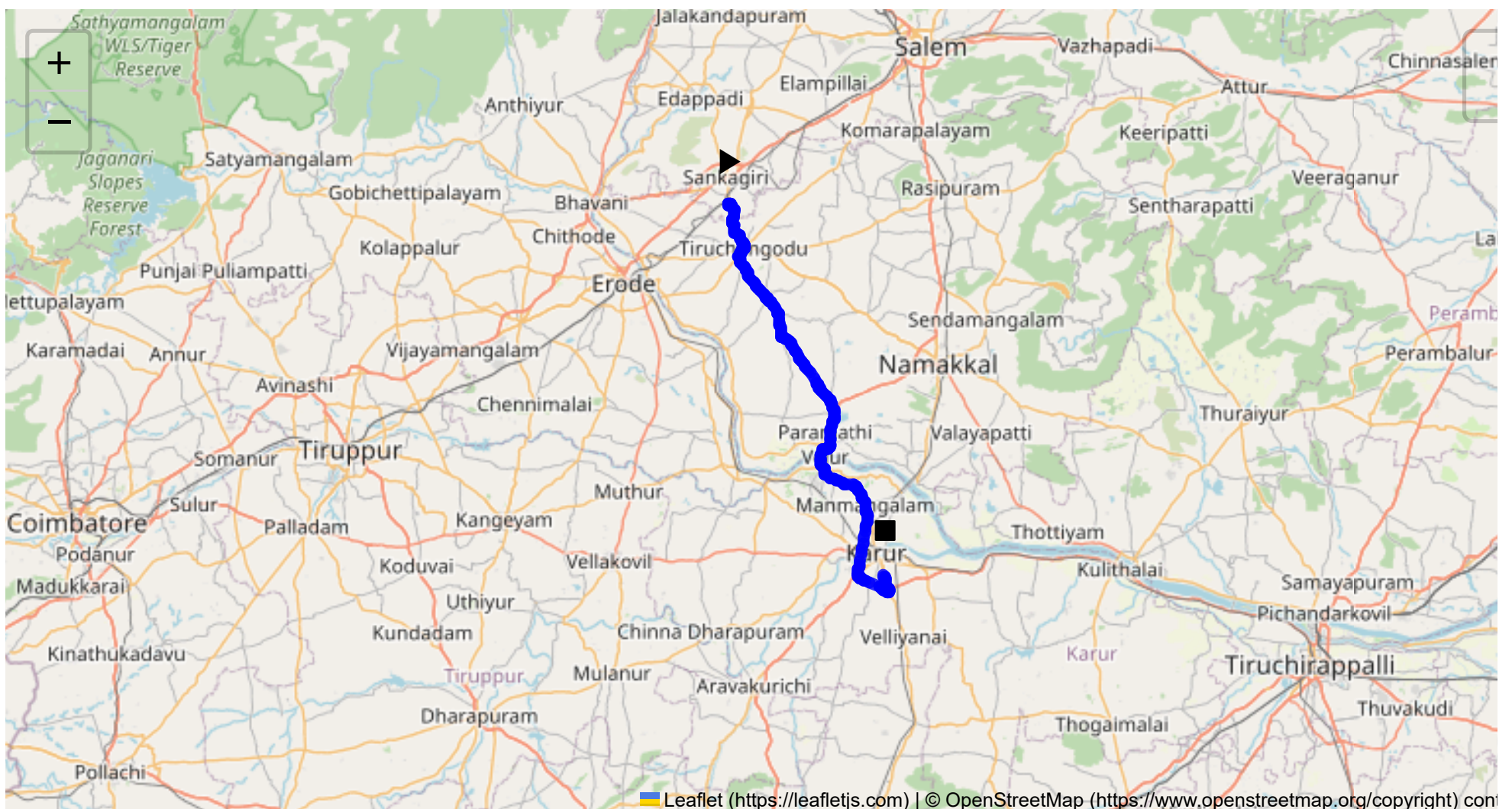
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 76.39 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours
Start: (11.4381, 77.8734)
End: (10.930081, 78.091648)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri, Tamil Nadu to Bharathidasan Nagar, Kaliyappanoor, Karur, Tamil Nadu covers approximately 76.39 kilometers. It passes through several key points including Puthur, Tiruchengode, Musalnaickenpalayam, Paramathi, Velur, and Karattuppalaiyam. The journey traverses primarily state highways and local roads, with a typical travel time of approximately 1.49 hours for heavy vehicles.

Typical Weather Conditions

Weather-Related Hazards:

- **Monsoon Season (June to September):** Heavy rains can cause waterlogging and reduce visibility. Roads can become slippery, especially in rural areas.
- **Summer (March to June):** High temperatures, leading to heat stress. Take precautions to avoid overheating in vehicles.

Traffic Patterns

Peak Hours and Congestion Zones:

- **Sangagiri to Tiruchengode:** Morning (8-10 AM) and evening (5-7 PM) see higher traffic volumes due to school and work commutes.
- **Tiruchengode to Karur:** Moderate traffic with potential congestion around local markets and commercial zones.

Road Quality and Infrastructure

Assessment:

- **General Road Condition:** Mostly adequate on state highways, but some local roads may have potholes or uneven surfaces.
- **Infrastructure:** Signage may be limited in rural stretches; ensure reliance on GPS for navigation.

Alternative Routes

For emergencies, consider using:

- **NH44/NH44:** Known for a higher standard of roadway, though can be crowded, offering access to larger cities and emergency services.
- **Local Bypasses:** Shortcuts around towns to avoid congested areas may be possible but should be checked for road conditions.

Regulations on Hazardous Material

Local Regulations:

- Permit required for transporting hazardous materials.
- Restrictions on transportation during night hours in certain areas to avoid accidents.

Historical Incidents

Notable Incidents:

- Occasional accidents involving heavy vehicles have been reported, often due to weather conditions or driver fatigue.

- No major incidents specifically involving hazardous materials noted recently.

Environmental Considerations

Sensitive Areas:

- Farmlands:** Ensure proper measures to prevent spillage.
- Water Bodies:** Crossings near rivers require extra caution to prevent environmental contamination.

Communication Coverage

Potential Dead Zones:

- Rural areas around Paramathi and Velur may have intermittent mobile network coverage.
- Use satellite communication if consistent contact is necessary.

Emergency Response Times

Estimated Response:

- Urban Areas (Sangagiri, Tiruchengode):** 15-30 minutes.
- Rural Areas (Velur, Paramathi):** 45-60 minutes due to distance from major emergency services.

Overall Risk Assessment Summary

The route presents moderate risks primarily due to weather-induced hazards and variable road quality in rural areas. Regular updates on weather and traffic conditions are emphasized. Drivers should maintain constant communication and be aware of emergency bypasses and protocols for hazardous material transport. With informed planning and adherence to safety measures, the route can be navigated safely.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43961, 77.87341	15 KM/Hr
4	Turn	High	11.44022, 77.87546	15 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16755, 78.01568	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.15579, 78.02169	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14480, 78.02291	15 KM/Hr
11	Turn	Medium	10.92922, 78.05481	30 KM/Hr
12	Turn	High	10.90784, 78.09916	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
6	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Manmangalam	11.0275504, 78.0646437	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.92958, 78.05452



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44022, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16755, 78.01568



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.15579, 78.02169



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.92922, 78.05481



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.90784, 78.09916
